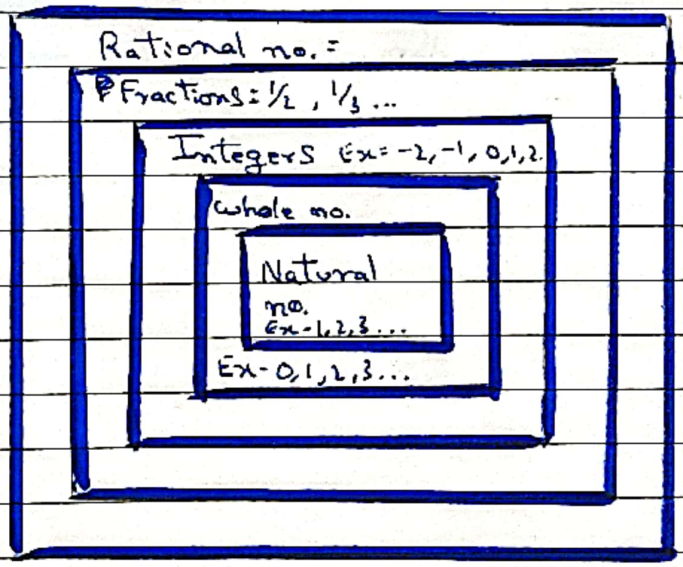
12/4/25

Real Numbers

- Real numbers are combination of rational and irrational numbers.

Real Numbers

Rational	Irrational
no. written in the form p/q, $q \neq 0$ and p & q are integers	no. which cannot be written in the form of p/q.



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- Examples of Rational no. $-\frac{1}{3}$, -6, 82, etc
- Examples of Irrational no. - $\sqrt{2}$, $1.572\ldots$

Decimal Expansion of Real Numbers

Rational no.

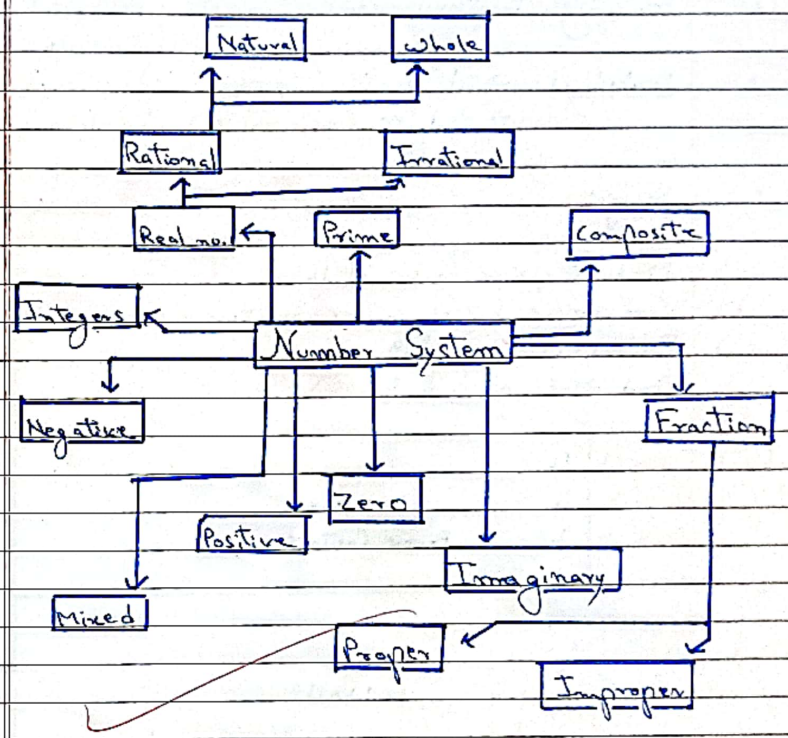
- Terminating, eg - $1/2$, $1/5$, $1/3$, $1/4$...
- Non-Terminating, eg - $1/3$, $5.\overline{76}$, etc
  repeating

Irrational no.

Non-Terminating, non-repeating
eg - $\sqrt{5}$, $7.89624\ldots$, etc

- If the remainder is zero, it means the no. is terminating.

Mind Map of Number System



- **Contents:**

① Fundamental Theorem of Arithematic

② Revisiting Irrational no.

- **Vocabulary:** → Arithematic (+, −, ×, ÷)
  → Real no. (Rational + Irrational no.)

- **Recap:**

Q1) Find a prime factors of 12
Q2) Find prime factors of 26 by tree diagram
Q3) Find LCM of 12 & 18
Q4 Find HCF of 12 & 18

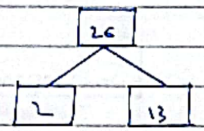
Ans 1

3	12
2	4
2	2
	1

$12 = 2 \times 2 \times 3$

Prime factors = 2, 3

Ans 2



$26 = 2 \times 13$

Ans 3

$12 = 2 \times 2 \times 3$
$18 = 2 \times 3 \times 3$
$LCM = 2 \times 3 \times 2 \times 3 = 36$

Ans 4

$12 = 2 \times 2 \times 3$
$18 = 2 \times 3 \times 3$
$HCF = 2 \times 3 = 6$

- **Fundamental Theorem of Arithematic:**

* It says every composite no. can be expressed as a product of prime numbers.
* Every composite no. can be break into prime numbers.
* Apart from the order, factorisation is unique.

for eg:-

$12 = 2 \times 2 \times 3$
$12 = 3 \times 2 \times 2$
$12 = 2 \times 3 \times 2$

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Exercise 1.1

1-i-Ans $140 = 2\times2\times5\times7 = 2^2\times5^1\times7^1$

ii-Ans $156 = 2\times2\times3\times13 = 2^2\times3^1\times13^1$

iii-Ans $3825 = 3\times3\times5\times5\times17 = 3^2\times5^2\times17^1$

iv-Ans $5005 = 5\times7\times11\times13$

v-Ans $7429 = 17\times17\times23$

2-i-Sol. → 26 and 91

$26 = 2\times13$

$91 = 7\times13$

HCF = 13

LCM = $2\times7\times13 = 182$

Product of two numbers = $26\times91 = 2366$

HCF × LCM = $13\times182 = 2366$

Hence, Product of two numbers = HCF × LCM

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ii-Sol. → 510 and 92

$510 = 2\times3\times5\times17$

$92 = 2\times2\times23$

HCF = 2

LCM = $2\times2\times3\times5\times17\times23 = 23460$

Product of two numbers = $510\times92 = 46920$

HCF × LCM = 2×23460

$= 46920$

Hence, product of two numbers = HCF × LCM

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iii-Sol. → 336 and 54

$336 = 2\times2\times2\times2\times3\times7$

$54 = 2\times3\times3\times3$

HCF = $2\times3 = 6$

LCM = $2^4\times3^3\times7 = 3024$

Product of two numbers = $336\times54 = 18144$

HCF × LCM = $6\times3024 = 18144$

Hence, product of two numbers = HCF × LCM

3-i-Sol. → 12, 15 and 21

$12 = 2^2\times3$

$15 = 3\times5$

$21 = 3\times7$

HCF = 3

LCM = $2^2\times3\times5\times7 = 420$

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ii-Sol. → 17, 23 and 29

$17 = 1\times17$

$23 = 1\times23$

$29 = 1\times29$

HCF = 1

LCM = $17\times23\times29 = 11339$

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iii-Sol. → 8, 9 and 25

$8 = 2\times2\times2$

$9 = 3\times3$

$25 = 5\times5$

HCF = 1

LCM = $2\times2\times2\times3\times3\times5\times5 = 1800$

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4-Solu HCF (306, 657) = 9

we know that, LCM × HCF = Product of two numbers

∴ LCM × HCF = 306 × 657

$LCM = \frac{306 \times 657}{HCF} = \frac{306 \times 657}{9}$

LCM = 22338

5-Solu if any number ends with the digit 0, it should be divisible by 10 or in other words, it will also be divisible by 2 and 5 as 10 = 2×5

Prime factorisation of $6^n = (2\times3)^n$

It can observed that 5 is not in the prime factorisation of 6^n.

Hence, for any value of n, 6^n will not be divisible by 5.

Therefore, 6^n cannot end with the digit 0 for any natural number n.

6-Solu Numbers are of two types - prime and composite. Prime numbers can be divided by 1 and only itself, whereas composite numbers have factors other than 1 and itself.

It can be observed that

$7\times11\times13+13 = 13(7\times11+1) = 13(78) = 13\times13\times6$

The given expression has 6 and 13 as its factors. Therefore, it is a composite number.

$\Rightarrow 7\times6\times5\times4\times3\times2\times1 + 5 = 5(7\times6\times4\times3\times2\times1+1)$

$= 5\times(1008+1)$

$= 5\times1009$

1009 cannot be factorised further. Therefore, the given expression has 5 and 1009 as its factors. Hence, it is a composite number.

7-Solu Time taken by Sonia = 18 minutes

Time taken by Ravi = 12 minutes

LCM of 12 and 18 = 2×2×3×3 = 36

Therefore, Ravi and Sonia will meet together at the starting point after 36 minutes.

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Exercise-1.2

1-Solu Let $\sqrt{5}$ be a rational number

$\Rightarrow \sqrt{5} = \frac{p}{q}$

Squaring on both sides

$\Rightarrow 5 = \frac{p^2}{q^2}$

$\Rightarrow q^2 = \frac{p^2}{5}$

$\therefore$ 5 divides p^2

$\therefore$ 5 divides p

$\Rightarrow 5q^2 = p^2$ —(1)

$\cancel{\Rightarrow 5q^2 = 5q}$

$\cancel{\Rightarrow 5q^2 = 5p}$

Let ~~5q = 4p~~ $5c = p$, for some integer c

$\Rightarrow 5q^2 = (5c)^2$ (from (1))

$\Rightarrow 5q^2 = 25c^2$

$\Rightarrow \frac{\cancel{5}q^2}{\cancel{25}5} = c^2$

$\Rightarrow$ 5 divides q^2

$\therefore$ 5 divides q

Hence, p and q have a common factor 5

This is a contradiction and our assumption is wrong. $\sqrt{5}$ is an irrational no.

2-Solu Let $3+2\sqrt{5}$ be a rational no.

$\Rightarrow 3+2\sqrt{5} = \frac{p}{q}$

$\Rightarrow 2\sqrt{5} = \frac{p}{q} - 3$

$\Rightarrow \sqrt{5} = \frac{p-3q}{2q}$

$\because \frac{p-3q}{2q}$ is a rational no.

$\Rightarrow \sqrt{5}$ is a rational no. which is a contradiction.

Our assumption is wrong. Hence, $3+2\sqrt{5}$ is irrational no.

3-i-Solu $\Rightarrow$ Let $\frac{1}{\sqrt{2}}$ be a rational no.

Rationalising

$\frac{1}{\sqrt{2}} = \frac{1}{2}\sqrt{2}$ is rational number.

$\sqrt{2}$ is rational, which is impossible because $\sqrt{2}$ is irrational.

Thus, our assumption is wrong

Hence, $\frac{1}{\sqrt{2}}$ is irrational.

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(ii) Sol: Lets $7\sqrt{5}$ be a rational no.

$\rightarrow 7\sqrt{5} = \frac{a}{b}$

$\rightarrow \sqrt{5} = \frac{a}{7b}$ which is rational

$\rightarrow \sqrt{5}$ is rational

which is a contradiction

$\therefore$ our assumption is wrong

Hence, $7\sqrt{5}$ is irrational

(iii) Sol: Let $6+\sqrt{2}$ be a rational no.

$\rightarrow 6+\sqrt{2} = \frac{a}{b}$

$\rightarrow \sqrt{2} = \frac{a}{b} - 6$

$\rightarrow \sqrt{2} = \frac{a-6b}{b}$ which rational.

$\sqrt{2}$ is rational

which is a contradiction.

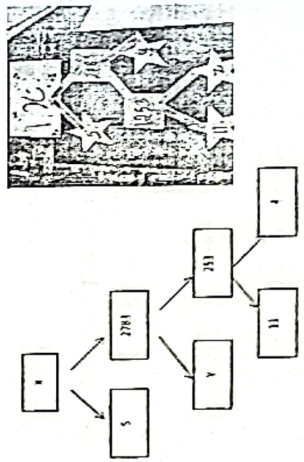
$\therefore$ Our assumption is wrong.

Hence, $6+\sqrt{2}$ is irrational.

D. If p and q are positive integers such that $p = ab^2$ and $q = a^3b$, where a , b are prime numbers, then the LCM (p, q) is_____.

a) ab b) a^2b^2 c) a^3b^2 d) a^3b^3

9) A Mathematics Exhibition is being conducted in your School and one of your friends is making a model of a factor tree. He has some difficulty and asks for your help in completing a quiz for the audience. Find x,y and z.



10. Prove that $5\sqrt{7} - 2\sqrt{7} + \sqrt{7}$ is an irrational number.

11. Prove that $\sqrt{p} + \sqrt{q}$ is irrational, where p,q are primes.

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